

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] MAY-JUNE 2014

Paper Code: ETME-202

Subject: Kinematics & Dynamics
of Machines

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions including Q.no.1 which is compulsory. Select one question from each Unit.

Q1 SHORT QUESTIONS:-

(5x5=25)

- (a) Discuss Inversions
- (b) Discuss transmissibility ratio
- (c) Explain the meaning of fixing torque
- (d) Explain Gyroscopic law
- (e) Explain degree of freedom

Unit-I

- Q2 In a crank and slotted lever quick return motion the distance between the centres is 160 mm and the length of the driving crank is 80 mm. Find the inclination with the vertical of the slotted lever in the extreme position and the ratio of time of cutting to time of return. The length of the slotted lever from the centre of the oscillation to its other end is 300 mm. Estimate also the stroke length if the line of the stroke passes through extreme position of the free end of the lever. (12.5)

- Q3 Two parallel shafts are to be connected by a pair of gears. The centre to centre distance between the axes of the shaft is approximately 600 mm. The velocity ratio between the shafts is to be 2.7. If module of mating gears is 7 mm. Find the number of teeth of the gears and the actual distance between the shaft axes. (12.5)

Unit-II

- Q4 In a reversed gear train, the arm A carries two wheels B and C and a compound wheel D-E. The wheel B gears with wheel E and the wheel C gears with wheel D. The number of teeth on wheel B, C and D are 75, 30 and 90 respectively. Find the speed and direction of wheel C when wheel B is fixed and the arm A makes 100 rpm clockwise. (12.5)
- Q5 The follower moves with uniform acceleration/retardation during ascent and with S.H.M. during decent. Draw the cam profile if the Least radius of cam = 30 mm, angle of ascent = 80 degree, angle of dwell between ascent and decent = 60 degree, angle of decent = 120 degree, roller diameter = 20 mm, lift of follower = 40 mm, offset of the roller = 15 mm. If the cam rotates 300 r.p.m. Then determine the maximum velocity and acceleration of the follower during ascent and decent. (12.5)

Unit-III

- Q6 (a) Explain Kinematics pair and also different types of Kinematics pair with suitable diagram and applications. (6.5)
(b) Explain the effect of gyroscopic couple on automobiles with suitable diagram and examples. (6)
- Q7 Find the balance masses required to balance all the revolving masses and two third of reciprocating masses of an inside cylinder uncoupled locomotive. The cylinders are 60 cm apart and the balance planes 150 cm apart. Mass of revolving part per cylinder is 525 kg, reciprocating parts per cylinder are of mass 375 kg, crank radius = 30 cm, and the mass centre's of the balance weights are at a radius of 75 cm. (12.5)

Unit-IV

- Q8 A shaft 300 mm diameter, 1.5 m long is simply supported at its end. It is subjected to isolated loads of 1 kN and 2 kN at distance of 400, 600, 800, and 1000 mm, from the left hand support respectively. Assuming the shaft density is 8 gram per cu. cm. Determine the natural frequency of the free transverse vibration of the shaft. Assume $E = 2 \times 10^5 \text{ MN/m}^2$. (12.5)
- Q9 (a) Explain the law of gearing with suitable diagram and also explain its applications. (6.5)
(b) Explain vibration analysis of SDOF systems with neat and clean diagram. (6)

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] MAY-JUNE-2015

Paper Code: ETAT 202

Subject: Theory of Machines
(Batch: 2013 onwards)

Time : 3 Hours

Maximum Marks : 75

Note: Attempt any five questions including question no.1 which is compulsory, select one question from each unit.

Q1. Short answer type questions:

(5x5=25)

- Explain transmissibility ratio.
- Describe function of flywheel.
- Explain damped forced vibration.
- Define Grublers rule for degree of freedom.
- Describe cycloidal motion of a cam.

Unit-I

- Q2. a) Describe the different types of Kinematic pairs with suitable diagram and applications. (6.5)
b) What is inversion, discuss the various types of inversions with neat diagram and examples. (6)
- Q3. A flywheel having a mass of 4 tonne has a radius of gyration 2m. What amount of energy this flywheel will store in it in changing its speed from 420 rpm to 462 rpm. (12.5)

Unit-III

- Q4. Two parallel shafts are to be connected by a pair of gears. The centre to centre distance between the axes of the shaft is to be approximately 600mm. The velocity ratio between the shaft is to be 2.7. If module of the mating gears is 7mm, find the number of teeth of the gears and the actual distance between the shaft axes. (12.5)
- Q5. The two shafts A and B in the same line, are geared together through an intermediate parallel shaft C. The wheel connecting A and C have a module of 4 and those connecting C and B a module of 9. The least number of teeth in any wheel being not less than 15. The speed of A is to be about but greater than 12 times the speed of B and the ratio at each reduction is same. Find suitable wheel and actual and the distance of shaft C and A and B. (12.5)

Unit-II

- Q6. With a data given below the follower moves with the uniform acceleration/retardation during ascent and with S.H.M. during descent. Draw the cam Profile Least radius of cam=30mm, Angle of ascent=80°, Angle of dwell between ascent and decent=60°, Angle of decent=120° Roller dia.=20mm, lift of follower=40mm, offset of follower=15mm. If the cam rotates at 300 r.p.m. then determine the maximum velocity and acceleration of the follower during ascent and decent. (12.5)
- Q7. The porter governor has four arms of equal length each of 200mm. The upper arms are pivoted on the axis of rotation of the spindle and the lowers at a distance of 25 mm from the axis. The radius of rotation of balls at a speed of 250 r.p.m. is 100mm. find the governor speed after the sleeve has lift 50mm. Also calculate the governor effort and power, if the mass of each ball is 2kg and mass of load placed on the sleeve is 20 kg. (12.5)

Unit-IV

- Q8. Three masses are attached to a shaft as follows: 10 kg at 90mm radius, 15 kg at 120mm radius and 9kg at 150mm radius. The masses are to be arranged so that the shaft is in complete balance. Determine the angular position of masses relative to 10 kg mass. All the masses are in the same plane. (12.5)
- Q9. In a single degree damped vibrating system, a suspended mass of 8kg makes 30 oscillations in 18 seconds. The amplitude decreases to 0.25 of the initial value after five oscillations. Determine:
(a) The stiffness of the spring.
(b) The logarithmic decrement.
(c) The damping factor.
(d) The damping co-efficient. (12.5)

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END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] MAY-JUNE 2016

Paper Code: ETAT-202**Subject: Theory of Machines****Time: 3 Hours****Maximum Marks: 75**

Note: Attempt any five questions including Q.no.1 which is compulsory. Select one question from each Unit. Assume missing data if any.

- Q1 Write short notes on: (5x5=25)
- Inversion of 3R-P chains
 - Cycloidal motion
 - Law of gearing
 - Forced vibration
 - Transmissibility ratio

Unit-I

- Q2 (a) Explain determination of velocity of a point on a link by instantaneous centre method. (5)
- (b) In a slider crank mechanism the length of crank. OB and connecting rod AB are 125 mm and 500 mm respectively. The centre of gravity, G of connecting rod is 275 mm from the slider A. The crank speed is 600 RPM clockwise. When the crank has turned 45° from the inner dead centre position, determine (i) velocity of the slider A (ii) Velocity of the point G (iii) Angular velocity of connecting rod AB. (7.5)

- Q3 (a) Explain Grubler rule for degree of freedom. (5)
- (b) The lengths of various links in a pin jointed four bar mechanisms are as follows.
 AB = 250 mm; BC = 500 mm; CD = 350 mm & AD (fixed) = 700 mm.
 Angle BAD = 60° . The crank AB rotates clockwise with angular velocity of 20.8 rad/sec and has angular acceleration of 500 rad/sec². Find (i) acceleration of the points B & G and (ii) angular acceleration of link BC. (7.5)

Unit-II

- Q4 (a) Draw the displacement, velocity and acceleration diagrams for a follower which it moves with SHM. Explain. (5)
- (b) A cam profile consist of two circular arcs of radii 24 mm and 12 mm, joined by straight lines, giving the follower a lift of 12 mm. The follower is a roller of 24 mm radius and its line of action is a straight line passing through the camshaft axis. When the cam shaft has a uniform speed of 500 RPM, find the maximum velocity and acceleration of the follower while in contact with the straight flank of the cam. (7.5)

- Q5 (a) Explain the effect of the gyroscopic couple on the reaction of the four wheels of a vehicle negotiating a curve. (5)
- (b) A ship is pitching through a total angle of 15° , the oscillation may be taken as simple harmonic and the complete period is 32 seconds. The turbine rotor has a mass of 6 tonnes, its radius of gyration is 450 mm and it is rotating of 2400 RPM. Calculate the maximum value of the gyroscopic couple set up by the rotor. What is the maximum angular acceleration to which the ship is subjected while pitching. (7.5)

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Unit-III

- Q6 (a) Explain with neat diagrams the following properties of gear teeth
(i) involute (ii) cycloidal. (6)
(b) When do you prefer epicyclic gear train? Derive the expression for velocity ratio of epicyclic gear train. (6.5)
- Q7 (a) Derive the expression for relation between the height of the governor and the angular speed of the balls of a Porter Governor. (5)
(b) With neat sketch explain the working of a Hartnell Governor. (7.5)

Unit-IV

- Q8 (a) Explain analytical method of balancing of several masses rotating in the same plane. (5)
(b) A simple cylinder reciprocating engine has speed 240 RPM, stroke 30 cm, weight of reciprocating parts 50 kg, weights of revolving parts at 15 cm radius is 37 kg. If $\frac{2}{3}$ rd of the reciprocating parts and all the revolving parts are to be balanced, find: (7.5)
(i) The balance weight required at a radius 40 cm
(ii) The residual unbalanced force, when the crank has rotated 60° from top dead centre.
- Q9 (a) Derive natural frequency of free transverse vibration. (6.5)
(b) Write a note on: (6)
(i) Excited vibration
(ii) Forced vibration
(iii) Natural vibration

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] MAY- JUNE 2017

Paper Code: ETAT-202

Subject: Theory of Machines

Time: 3 Hours

Maximum Marks: 75

Note: Attempt any five questions including Q.No1 which is compulsory.
Select one question from each unit. Assume missing data if any.

- Q1 Write short notes on following topics:- (10×2.5=25)
- Define mechanisms and degree of freedom.
 - Draw a neat sketch of a cam with offset follower and label its different components.
 - Differentiate between machine and structure.
 - Define complete and incomplete constraints in lower and higher pairs.
 - Differentiate between flywheel and governor.
 - Draw turning moment diagram for multi cylinder engine.
 - Define fluctuation of energy and co-efficient of fluctuation of speed.
 - Write two advantages and two disadvantages of involute profile over cycloidal profile.
 - Define damped free vibration and critical damping coefficient.
 - Explain logarithmic decrement and magnification factor. Also write their formula.

UNIT-I

- Q2 (a) Sketch and explain any two inversions of single slider crank chain. (5)
(b) In a pin jointed four bar mechanisms ABCD, the lengths of various links are as follows: AB=25 mm, BC = 87.5 mm, CD = 50 mm and AD = 80 mm.
The link is fixed and the angle BAD=135°. If the velocity of B is 1.8 m/s in the clockwise direction, find velocity of the midpoint of BC. (7.5)
- Q3 Locate all the instantaneous centers of the slider crank mechanism as shown in fig-1. The lengths of crank OB and connecting rod AB are 100 mm and 400 mm respectively. If the crank rotates clockwise with an angular velocity of 10 rad/s, find (i) Velocity of the slider A and (ii) Angular velocity of the connecting rod AB. (12.5)

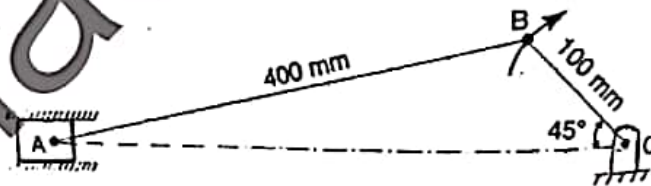


Fig-1

UNIT-II

- Q4 A cam operating a knife-edged follower has the following data:
- Follower moves outwards through 40 mm during 60° of cam rotation.
 - Follower dwells for the next 45°.
 - Follower returns to its original position during next 90°.
 - Follower dwells for the rest of the rotation.

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P.T.O.

The displacement of the follower is to take place with simple harmonic motion during both the outward and return strokes. The least radius of cam is 50 mm. Draw the profile of the cam when the axis of the follower passes through the cam axis. If the cam rotates at 300 rpm, determine maximum velocity and acceleration of the follower during the outward stroke and the return stroke. (12.5)

- Q5 The turbine rotor of a ship has a mass of 3500 kg. It has a radius of gyration of 0.45 m and a speed of 3000 rpm clockwise when looking from stern. Determine the gyroscopic couple and its effect upon the ship: (12.5)
- When the ship is steering to the left on a curve of 100 m radius at a speed of 36 km/h.
 - When the ship is pitching in a simple harmonic motion, the bow falling with its maximum velocity. The period of pitching is 40 seconds and the total angular displacement between the two extreme positions of pitching is 12° .

UNIT-III

- Q6 (a) Draw epicyclic gear train. In an epicyclic gear train, an arm carries two gears A and B having 36 and 45 teeth respectively. If the arm rotates at 150 r.p.m. in the anticlockwise direction about the centre of the gear A which is fixed, determine the speed of gear B. If the gear A instead of being fixed, makes 300 r.p.m. in the clockwise direction, what will be the speed of gear B? Explain why tabular method. (8)
- (b) Define law of gearing. Draw neat sketch of helical and worm gear. (4.5)
- Q7 Define and classify governor and explain any one type of governor. The flywheel of a steam engine has a radius of gyration of 1 m and mass 2500 kg. The starting torque of the steam engine is 1500 N-m and may be assumed constant. Determine (i) the angular acceleration of the flywheel and (ii) the kinetic energy of the flywheel after 10 seconds from start. (12.5)

UNIT-IV

- Q8 Differentiate between static and dynamic balancing. A, B, C and D are four masses carried by a rotating shaft at radii 100, 125, 200 and 150 mm respectively. The planes in which the masses revolve are spaced 600 mm apart and the mass of B, C and D are 10 kg, 5 kg and 4 kg respectively. Find the required mass A and the relative angular setting of the four masses so that the shaft should be in complete balance. (12.5)
- Q9 Explain free and forced vibration with proper example. The following data are given for a vibratory system with viscous damping: mass=2.5 kg, spring constant=3N/mm and the amplitude decreases to 0.25 of the initial value after five consecutive cycles. Determine the damping coefficient of the damper in the system. (12.5)

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Exam Roll No.

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] MAY-JUNE 2018

Paper Code: ETAT-202

Subject: Theory of Machines

(Batch 2013 Onwards)

Time: 3 Hours

Maximum Marks: 75

Note: Attempt any five questions including Q no.1 which is compulsory.
Assume suitable missing data if any.

- Q1 Short answer type questions:- (5x5=25)
- (a) Explain Klein's construction with neat sketch, write their applications.
 - (b) Discuss different types of kinematics pairs.
 - (c) Discuss the effect of gyroscopic couple on ships and aircrafts.
 - (d) Discuss compound gear train and epicyclic gear train with neat sketch.
 - (e) Describe transmissibility ratio, balancing of revolving masses and balancing of reciprocating engine.
- Q2 (a) State and prove the laws of gearing. Show that involute profile satisfies the condition for correct gearing. (6.5)
- (b) Derive an expression for minimum numbers of teeth required on a pinion to avoid interference when it gears with a rack. (6)
- Q3 A single cylinder motor cycle engine runs at 1500 r.p.m. and has a stroke of 20 mm. The piston has a mass of 0.3 kg and mass of connecting rod is 0.6 kg. The revolving masses at crank pin are 0.4 kg. If all revolving and 65% of reciprocating masses are balanced by placing a balance mass at 30 mm radius. Find the balance mass and also draw a graph of resultant unbalanced force for crank angles varying from 0 to 180 degree. Assume that two-third mass of connecting rod acts at crank pin. (12.5)
- Q4 In a simple watt governor the length of upper arm is 10 mm and its inclination to the vertical is 30 degree. Estimate the percentage change in speed for a rise in the level of balls by 20 mm. (12.5)
- ✓Q5 Draw a profile of cam operating knife edge follower from the following data.
- (a) Follower to move outward through 40 mm during 60 degree of cam rotation.
 - (b) Follower to dwell for the next 45 degree.
 - (c) Follower to return to its original position to next 90 degree.
 - (d) Follower to dwell for the rest of cam rotation.
- The displacement of the follower is to take place with S.H.M during both outward and return strokes. The least radius of the cam is 50 mm. If the cam rotates at 300 rpm., determine the maximum velocity and acceleration of the follower during the outward stroke and the return stroke. (12.5)
- ✓Q6 Explain the concept of flywheel. A flywheel having a mass of 4 tones has a radius of gyration of 2 m. What amount of energy this flywheel will store in it in changing its speed from 420 to 462 r.p.m. (12.5)
- ✓Q7 (a) Explain the inversions of 3R-P, 2R-2P chains with suitable examples and diagrams. (6.5)
- (b) Explain the Grublers rule for degree of freedom with suitable example and applications. (6)
- ✓Q8 A steel shaft 60 mm dia and 2.5 m long is fixed horizontally at one end. It is loaded with a load of 300 N at its free end. Calculate the frequency of its free transverse vibrations if $E = 2.5 \times 10^5 \text{ MN/m}^2$. (12.5)
